

NeuroScript

AI-Based Handwriting Analysis for Neurological Detection

Leveraging machine learning and explainable AI to enable early, non-invasive screening for OCD and ADHD through handwriting.

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An illustration of a person's hands writing in a green spiral-bound notebook on a wooden desk. The person is wearing a green sweater and a tan jacket. The notebook has the words "Ab dd do in" and "a kix m m" written in cursive. On the desk, there is a red pencil, a white eraser, a blue pen, and a small potted plant. The scene is lit with warm, golden light, suggesting a sunny day.

The Problem with Today's Diagnosis

The Challenge

OCD and ADHD are frequently diagnosed late — often after years of unrecognised symptoms. **Current clinical methods rely heavily on:**

- Subjective assessment
- Structured interviews
- Behavioural observation

These methods are:

- Time-intensive
- Inconsistent across practitioners

Additional Challenges:

- Lack of localized (Pakistani) datasets
- No systems using real-time handwriting input

The NeuroScript Solution

An AI-powered handwriting analysis system that enables early, non-invasive neurological detection.

Key Features:

- Uses Pakistani handwriting dataset
- Accepts real-time input via stylus/finger
- Provides instant predictions with explanations
- Designed for clinicians, students, and researchers

Why Handwriting? Closing the Diagnostic Gap

Traditional Diagnosis

Clinical assessments vary by practitioner, environment, and patient state — introducing bias and inconsistency into every evaluation.

Behavioural Data

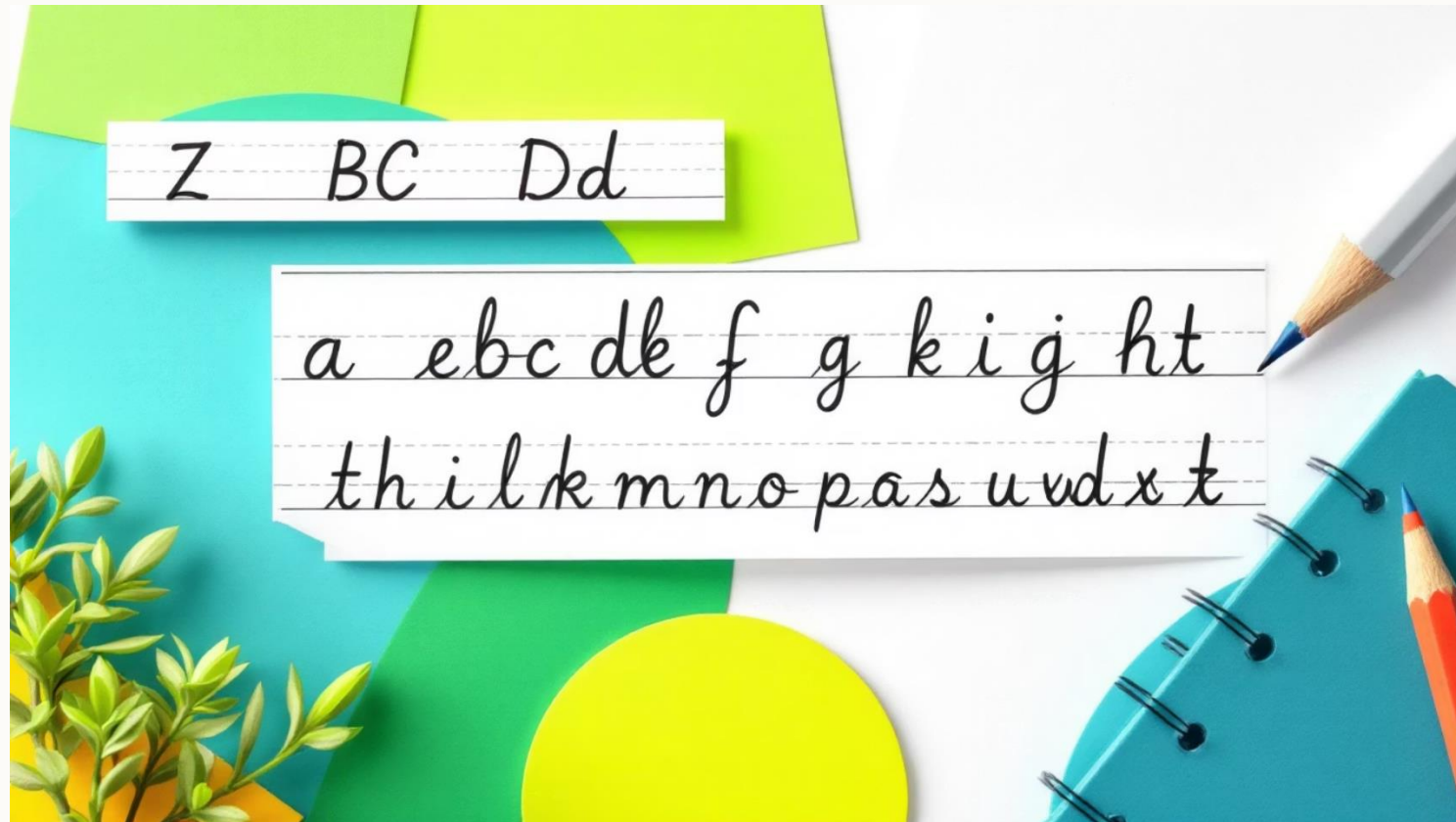
Handwriting encodes rich neuromotor signals — spacing, letter formation — that are rarely leveraged in diagnostic workflows.

Lack of Explainability

Existing AI tools offer predictions without justification, limiting clinician trust and adoption in real-world practice.

The Gap: There is no widely adopted system that combines automated detection with interpretable, clinician-friendly explanations.

Dataset & Preprocessing



Data Sources

Dataset (Pakistani-Based)

Handwriting samples collected from **Pakistani students**

Including English writing.

Covers multiple age groups

Data collected using:

- Paper-based samples (digitized)
- Digital input via stylus/finger

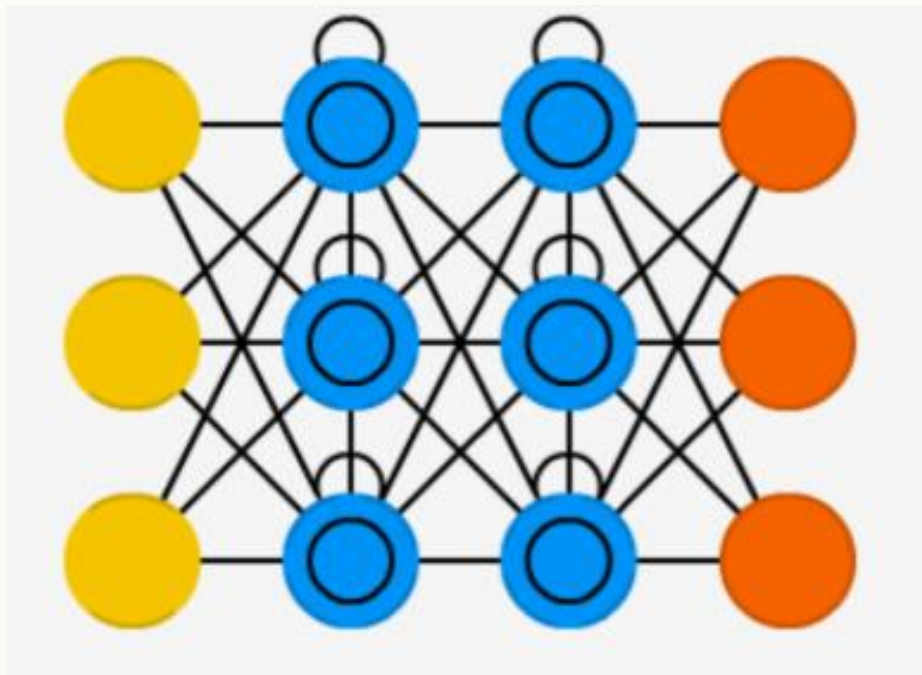
Labeled into categories:

- Normal
- Potential ADHD/OCD indicators

Why It Matters

- Handwriting reflects neuromotor behavior linked to brain function
- Disorders like ADHD affect:
 - Writing consistency
 - Spacing
 - Stroke patterns
- Dysgraphia is strongly correlated with ADHD, making handwriting a valuable indicator
- Using a Pakistani dataset improves:
 - Accuracy
 - Cultural relevance
 - Real-world applicability

Predictive Modelling



Long Short-Term Memory (LSTM)

Our system utilizes a **Long Short-Term Memory (LSTM)** network to analyze stylus-based handwriting data.

LSTM is a type of recurrent neural network designed to process **sequential and time-dependent data**, making it well-suited for capturing dynamic handwriting behavior.

The model learns patterns from:

- **Stroke order**
- **Writing speed**
- **Pen movement and timing**

Unlike traditional image-based models, LSTM enables deeper analysis of neuromotor patterns, which are critical indicators in detecting conditions like ADHD and OCD.

This approach allows the system to move beyond static handwriting analysis and leverage real-time behavioral signals for more accurate and meaningful predictions.

Explainable AI — Building Clinician Trust

AI is transforming the way we create pictures and content., Some people learn quickly, adapt well, and then use AI to improve their work - but should we replace human creativity with AI tools?

Why Explainability Matters

Predictions without justification are insufficient in clinical settings.

NeuroScript integrates Explainable AI (XAI) to ensure that every prediction is:

- Transparent
- Interpretable
- Clinically meaningful

This is especially important when working with Pakistani patient data, where trust and clarity are essential for real-world adoption.

What the System Explains

The model highlights key handwriting features such as:

- Irregular spacing
- Inconsistent letter formation
- Stroke variation
- Writing speed and movement patterns (from stylus input)

Directly supporting clinical decision-making

SHAP (SHapley Additive exPlanations)

- Quantifies each feature's contribution to the prediction
- Identifies which handwriting traits influenced the result most

LIME (Local Interpretable Model-Agnostic Explanations)

- Explains individual predictions
- Shows how the model behaves for a specific handwriting sample

Tools & Technologies



Python

Core development language powering the entire pipeline.



SHAP & LIME

Explainability frameworks providing feature attribution and local model interpretations.



TensorFlow

Deep learning framework used to build and train the LSTM model for sequential handwriting analysis.



Streamlit

Front-end frameworks for deploying the interactive, web-based diagnostic interface.



Numpy & Pandas

Used for handling and preprocessing time-series stroke data (coordinates, timing, speed).

Input

Preprocess

Classify

Explain

